PH-UY 2023: ELECTRICITY, MAGNETISM, & FLUIDS

MIDTERM EXAM Name _____

3/29/22 Univ ID N_____ Section _____



BEFORE YOU DO ANYTHING ELSE….

1) **Print your full name neatly on your answer sheet in the Name area.**
2) **Write your course AND SECTION in the Subject area.**
3) **Write your N# in the first eight boxes under "STUDENT ID NUMBER".**
4) **Make sure you use #2 pencil. Pen will not register when graded!**

1. A wooden raft has a density of 520 kg/m^3 and a volume of 2.2 m^3. What is the maximum number of 75-kg sacks that can be loaded onto this raft without it sinking in water?

a) 14 b) 11 c) 16 d) 20 e) 23

2. After a hole of a 1.4-inch diameter was punched in the hull of a yacht 40 cm below the waterline, water started pouring inside. At what rate is water flowing into the yacht? (1 in = 2.54 cm, 1 L = 10^{-3} m^3)

a) 2.78 L/s b) 2.31 L/s c) 3.68 L/s
d) 3.41 L/s e) 3.11 L/s

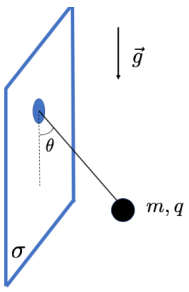
3. A 30-m-wide dam built across a river can withstand a total force of water of 4.3×10^7 N. Assuming the dam is rectangular, what is the maximum level (height) of the water it can withstand?

a) 12.1 m b) 14.8 m c) 13.2 m d) 20 m e) 17.1 m

4. Two parallel rectangular flat metal sheets with very large areas are 2 mm apart. The sheets are identical except that they carry equal but opposite charges distributed essentially uniformly on each sheet. If the potential difference between the sheets is 10 volts, then the surface charge density on the positively charged plate is closest to (Note: 1 nC = 10^{-9} C)

a) 26.6 nC/m^2 b) 35.4 nC/m^2 c) 44.3 nC/m^2
d) 53.1 nC/m^2 e) 62.0 nC/m^2

5. A small bob of charge q = 5.20 μC (1 μC = 10^{-6} C) is attached by an insulated neutral string of length 75 cm to a very large wall with a uniform positive surface charge density σ = 3.54 μC/m^2. The bob is in static equilibrium while the string makes an angle θ = 30° with the wall (gravity points toward the bottom of the page). What is the weight of the bob?



a) 1.20 N b) 1.80 N c) 2.40 N d) 3.00 N e) 3.60 N

6. Two point charges Q_1 = + 8 Q and Q_2 = - 15 Q are held fixed at the points (0,a) and (a,0), respectively. The magnitude of the net electric field at the origin is $\frac{1}{4\pi\epsilon_0}\frac{Q}{a^2}$ times

a) 10 b) 13 c) 15 d) 17 e) 25

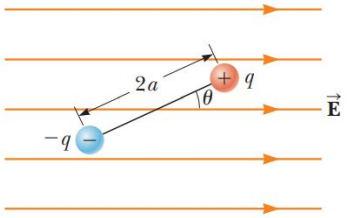
7. The y-axis carries a uniform linear charge density of -30 nC/m, and there is a 4.0 nC point charge at the point (4 cm, 0 cm, 0 cm) as well as a -12 nC point charge at the point (6 cm, 0 cm, 0 cm). What is the electric flux through a spherical surface of radius 5 cm centered at the origin?

a) 339 N·m^2/C b) 452 N·m^2/C c) 526 N·m^2/C
d) 113 N·m^2/C e) 226 N·m^2/C

8. A point charge of -3 nC is at the center of a spherical conducting shell of an 8-cm outer radius carrying a 9 nC charge. What is the surface charge density on the outer surface of the shell?

a) 81.2 nC/m^2 b) 37.3 nC/m^2 c) 74.6 nC/m^2
d) 62.2 nC/m^2 e) 49.7 nC/m^2

9. An electric dipole in a uniform horizontal electric field of magnitude 2.5 x 10^5 N/C is currently at rest and makes an angle θ = 17° with the E field. The charges have equal magnitudes q = 3.0 μC and are separated by a fixed distance 2a = 2.0 x 10^{-8} m. When released, the dipole swings clockwise about an axis through its center, with no friction or other damping, until it next comes to rest. At that moment, what angle will the dipole make with the E field?



a) 11° b) 14° c) 17° d) 21° e) 24°

10. An electron (charge -1.60 x 10^{-19} C and mass 9.11 x 10^{-31} kg) is initially at rest at the center of a uniformly charged ring of radius 3 mm and linear charge density λ = -4.07 nC/m. The electron is nudged slightly and starts accelerating away from the ring along its central axis. Which of the following is closest to the electron's speed by the time it is very far from the ring?

a) 5.0 x10^6 m/s b) 6.0 x 10^6 m/s c) 7.0 x 10^6 m/s
d) 8.0 x 10^6 m/s e) 9.0 x 10^6 m/s